Juan David Muñoz-Bolaños
Rechenauerstraße 42, Innsbruck, Austria (Tirol)
+43 676 9115145
juan.munoz@i-med.ac.at

TTControl S.r.l.
Hiring Team
Julius-Durst Straße 66
I-39042 Brixen / Bressanone, Italy

Subject: Application for Software Developer – Computer Vision and Robotics (m/f/d)

Dear Hiring Team,

As a physicist and computer vision engineer with hands-on C++/Python experience in real-time hardware control and FPGA-adjacent embedded systems, I am applying for the Software Developer role at TTControl in Brixen. My goal: make machines smarter through reliable, precisely engineered perception.

During my Ph.D. at the Medical University of Innsbruck, I built automated C++/Python pipelines for optical metrology and hardware integration, from design and specification through to testing and documentation. At IPHT Jena, I applied supervised ML to hyperspectral image data for automated classification. At INTECOL S.A.S., I deployed Halcon-based vision pipelines for high-speed industrial inspection, resolving critical production failures through custom illumination and optics design.

Originally from Colombia, I have been working in Innsbruck for the past four years as a researcher at the Medical University, with a certified B2 German level and full integration into DACH working culture. With my Ph.D. completing in May, I am available to join your team in Brixen shortly thereafter. I adapt quickly to new toolchains, including embedded Linux and ROS 2 environments, with the help of modern AI development tools.

Sincerely,

Juan David Muñoz-Bolaños
Optical and Computer Vision Engineer
Medical University of Innsbruck

Juan David Muñoz-Bolaños

Optical and Computer Vision Engineer

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

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juan.munoz@i-med.ac.at | LinkedIn Profile

PROFESSIONAL SUMMARY

Computer Vision and Optical Engineer with a strong foundation in C++/Python software development and real-time hardware control. Proven track record deploying machine vision and AI/ML solutions (including JAX for GPU acceleration) in both research and industrial environments, with experience in Labview FPGA, hardware-integrated perception pipelines, and strong problem-solving skills.

TECHNICAL COMPETENCIES

C++/Python and Real-Time Systems
Extensive experience developing real-time control software in C++ and Python, interfacing with hardware and managing complex embedded system architectures. Fast learner on new toolchains with modern AI-assisted development.

FPGA and Embedded Systems
Experience with Labview FPGA and real-time control architectures in optical instrumentation. Motivated to deepen expertise in Yocto, Buildroot, and ROS 2 production environments.

Machine Vision and AI Integration
Hands-on deployment of vision pipelines using Halcon and OpenCV. Applied JAX for GPU-accelerated computing and supervised ML to hyperspectral data. Strong interest in integrating AI/CV models on embedded systems.

Problem Solving and System Integration
Driven by strong analytical and problem-solving skills. End-to-end R and D experience from specification and design through testing and qualification, coordinating multi-component systems.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck — Innsbruck, Austria
PhD Researcher – Computer Vision and Embedded Systems
Jan 2022 – Present

- **Real-Time Software Architecture:** Designed and implemented end-to-end C++/Python software for automated real-time control of complex optical hardware, owning the full lifecycle from requirements to deployment. Experience with FPGA-adjacent control loops and fast signal acquisition pipelines.
- **Hardware/Software Integration:** Integrated adaptive optical components, precision motion stages, and scientific cameras into a unified automated system, with rigorous calibration and testing protocols.
- **Perception Pipeline Development:** Built automated wavefront sensing and phase retrieval pipelines applying signal processing and numerical algorithms to high-fidelity image data.
- **Cross-functional Collaboration:** Coordinated with researchers and hardware vendors (Thorlabs, Edmund Optics) ensuring reliable delivery of integrated systems for peer-reviewed publications.

Leibniz Institute of Photonic Technology (IPHT) — Jena, Germany
R and D Researcher – AI and Photonics
Jun 2020 – Dec 2021

- **AI/ML Integration:** Applied supervised machine learning to hyperspectral image data for automated tissue classification, bridging algorithmic research with practical laboratory implementation.
- **Systems Prototyping:** Designed and built a dispersive fiber probe Raman imaging spectrometer; calibrated point-scanning setups for clinical applications.

INTECOL S.A.S. — Medellín, Colombia
Junior Machine Vision System Engineer
Mar 2018 – Mar 2019

- **Industrial Vision Deployment:** Designed and deployed Halcon-based image processing pipelines for high-speed manufacturing inspection, resolving critical production failures through custom illumination and optics.
- **System Architecture:** Managed selection of cameras, sensors, and illumination, leading the full implementation of optical inspection setups on active production lines.

EDUCATION

Medical University of Innsbruck Austria
Ph.D. in Adaptive Optics & Microscopy *Jan 2022 – Present*

Friedrich Schiller University Jena Germany
M.Sc. in Photonics *2019 – 2021*

University of Cauca Colombia
B.Sc. in Physics Engineering *2012 – 2018*

PUBLICATIONS & AWARDS

- **Awards and Selections:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026 for best Colombian professionals.
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte Medical University of Innsbruck
Head of Institute for Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Dr. Christoph Krafft Leibniz Institute of Photonic Technology
Group leader: Spectroscopy Imaging
christoph.kraft@leibniz-ipht.de

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LEIBNIZ-INSTITUT für PHOTONISCHE TECHNOLOGIEN

AGREEMENT FOR GUESTS

to carry out a practical training

with

Mr. **Juan David Munoz Bolanos** **Date of birth: 1994-06-04**

temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** **(guest),**

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2021-01-01 up to 2021-09-30 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolanos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 13 November 2020

F. Sondermann

Prof. J. Popp	F. Sondermann	Juan David Munoz Bolanos
Chairman	Executive Member	Guest
of Leibniz-IPHT	of Leibniz-IPHT	

Leibniz | ipht
LEIBNIZ-INSTITUT für PHOTONISCHE TECHNOLOGIEN

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Mr.	**Juan David Munoz Bolanos**	**Date of birth: 1994-06-04**
temporarily living at	**Emil-Wölk-Str. 7, 07747 Jena**	**(guest),**

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

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Jena, 18 May 2020

F. Sondermann

Juan D. M.

Prof. J. Popp Chairman of Leibniz-IPHT	F. Sondermann Executive Member of Leibniz-IPHT	Juan David Munoz Bolanos Guest

INTECOL
OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos ($1,500,000) plus three hundred thousand pesos ($300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL
NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO
Administrative and Financial Director

FRIEDRICH-SCHILLER-UNIVERSITÄT JENA
PHYSIKALISCH-ASTRONOMISCHE FAKULTÄT

URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021

Friedrich - Schiller - Universität Jena
2

Prof. Dr. Christian Spielmann	Prof. Dr. Silvana Botti
Dekan	*Vorsitzende des Prüfungsausschusses*

FRIEDRICH-SCHILLER-UNIVERSITÄT JENA

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20 DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC OF COLOMBIA.

UNIVERSIDAD DEL CAUCA

ON BEHALF OF THE

REPUBLIC OF COLOMBIA

AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION

CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS

CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,

GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396

RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY

THE DEAN OF THE FACULTY

THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garces
TRADUCTOR E INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINJUSTICIA 1997

OFFICIAL TRANSLATION No. 0439

20 DECEMBER 2018

CERTIFICATE

Juan Munoz Bolanos

has participated in the

ZEISS European Summer School

Lithography Optics & Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Dr. Thomas Marschner
Program Manager Funding Projects

Dr. Thomas Stammler
CTO & Head of Product Strategy

ZEISS

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994
in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden

Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA

ÖSD ist Mitglied von

ALTE

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994
in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum
BFI Tirol Bildungs GmbH
in Innsbruck / Österreich

bestanden

österreich ösd INNSBRUCK schweiz deutschland

Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA

Europäisches Siegel für innovative Spracheninitiativen

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REPUBLICA DE COLOMBIA

PASAPORTE
PASSPORT

Tipo / *Type*	Cod. país / *Country code*	Pasaporte Nº / *Passport No.*
P	COL	AV410884

Apellidos / *Surname*
MUÑOZ BOLAÑOS

Nombres / *Given names*
JUAN DAVID

Nacionalidad / *Nationality*
COLOMBIANA

Fecha de nacimiento / *Date of birth*
04 JUN/JUN 1994

Núm. personal / *Personal No.*
CC1061772278

Sexo / *Sex*
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Lugar de nacimiento / *Place of birth*
LA CRUZ COL

Fecha de expedición / *Date of issue*
11 DIC/DEC 2018

Autoridad / *Authority*
G. ANTIOQUIA

Fecha de vencimiento / *Date of expiry*
10 DIC/DEC 2028

Firma del titular / *Holder's signature*

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SONSTIGE ANGABEN
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AUT
AUFENTHALTSTITEL
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NACHNAMEN Vornamen
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Juan David
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GEBURTSDATUM
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KARTE GÜLTIG BIS
16 05 2026
ART DES AUFENTHALTSTITELS
NIEDERLASSUNGSBEWILLIGUNG
SONSTIGE ANGABEN
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